

Calculus Homework – Continuity of real functions

Exercise 1. Investigate for continuity of the following functions!

- (1) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := x^k$, where $k \in \mathbb{Q}, k \geq 0$ is fixed,
- (2) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := 1$ if $x \in \mathbb{Q}$ and $f(x) := 0$ if $x \in \mathbb{R} \setminus \mathbb{Q}$,
- (3) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := |x|$,
- (4) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := x + 1$ if $x \leq 0$ and $f(x) := x$ if $x > 0$,
- (5) $f : \mathbb{R} \rightarrow \mathbb{R}$, $f(x) := x^4 + 3x + 7$.

Exercise 2. Let $f, g : \mathbb{R} \rightarrow \mathbb{R}$ be continuous functions such that $f(x) = g(x)$ if $x \in \mathbb{Q}$. Show that $f(x) = g(x)$ holds for all $x \in \mathbb{R}$!

Exercise 3. Let $f : \mathbb{R} \rightarrow \mathbb{R}$ be a continuous function and $p \in \mathbb{R}$ such that $f(p) > 0$. Show that there exists $\varepsilon > 0$ such that $f(x) > 0$ holds whenever $x \in]p - \varepsilon, p + \varepsilon[$!

Exercise 4. Calculate the discontinuities of the following functions and also determine their type! Calculate the one-sided limits at the discontinuities and at $-\infty$ and $+\infty$ as well!

- (1) $f(x) := \frac{x^2 + 6x - 7}{x^2 + 4x - 21}$, (2) $f(x) := \frac{3x^2 - 12}{x + 2}$, (3) $f(x) := \frac{2x^2 - 14x - 16}{x^2 + 2x + 1}$,
- (4) $f(x) := \frac{x^2 - 3x - 10}{x^2 - 3x - 18}$, (5) $f(x) := \frac{x^2 - 2x + 1}{x^2 - x + 2}$, (6) $f(x) := \frac{3x + 1}{6x^2 - x - 1}$
- (7) $f(x) := \frac{x^2 - 6x + 8}{x^2 - 2x - 8}$, (8) $f(x) := \frac{x^2 + 3x + 1}{2x + 6}$, (9) $f(x) := \frac{x^2 + 7x - 18}{x^2 + 1}$

Exercise 5. Give the value of $a \in \mathbb{R}$ such that $f : \mathbb{R} \rightarrow \mathbb{R}$ is continuous on $\mathbb{R}$!

- (1) $f(x) := \begin{cases} 2x + 1 & \text{if } x < 0, \\ a & \text{if } x = 0, \\ (x + 1)^2 & \text{if } x > 0, \end{cases}$
- (2) $f(x) := \begin{cases} \frac{3x^2 + 12x + 12}{x + 2} & \text{if } x \neq -2, \\ a & \text{otherwise.} \end{cases}$